

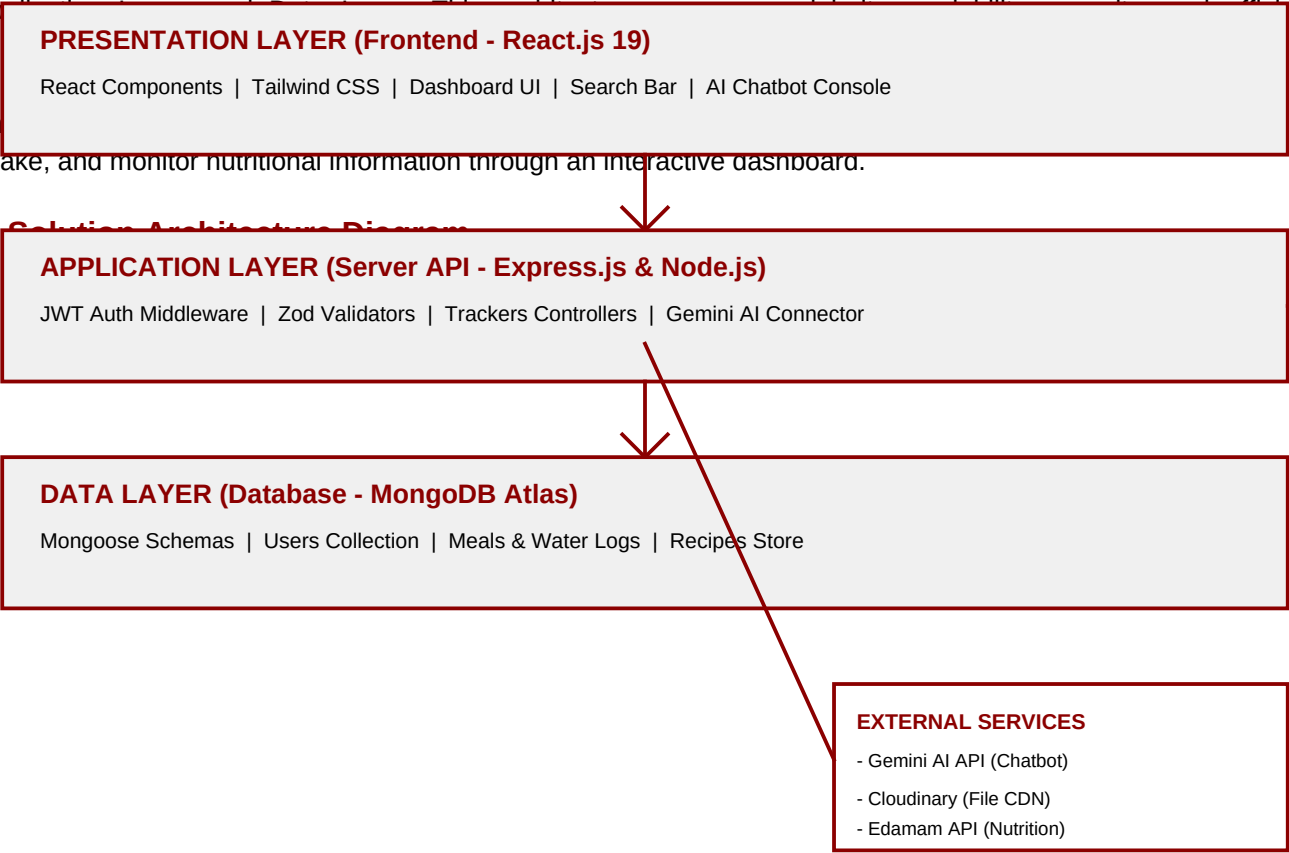
Project Design Phase: Solution Architecture

Date	7 July 2026
Team ID	
Project Name	Nutrition Assistant
Maximum Marks	4 Marks

1. Introduction

The Nutrition Assistant is a web-based application developed using the MERN Stack (MongoDB, Express.js, React.js, and Node.js). The system follows a three-tier architecture consisting of the Presentation Layer, Application Layer, and Data Layer. The Presentation Layer (Frontend) is responsible for user interface, data input, and output. The Application Layer (Server API) handles business logic, authentication, and data processing. The Data Layer (Database) stores user information, meal logs, and recipes. The system is designed to help users track their nutritional intake, and monitor nutritional information through an interactive dashboard.

2. Solution Architecture Diagram



3. Architecture Components

- **Presentation Layer (Frontend):** Developed using React.js and Tailwind CSS for visual interfaces, responsive dashboards, and interactive tracking inputs.
- **Application Layer (Backend):** Node.js and Express.js REST server. Manages user sessions (JWT), coordinates business routes, calculates caloric totals, and routes calls to cloud API utilities.

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- **Data Layer (Database):** MongoDB document storage mapping schemas. Manages persistent data queries securely.

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4. Data Flow

- **User Actions:** The client application triggers REST routes using the Axios API client.
- **Server Parsing:** Express.js validates fields using Zod validator schemas, extracts JWT details, and triggers corresponding controllers.
- **Database Operations:** Mongoose models query collections in MongoDB Atlas, returning results securely to the backend application.
- **Client Rendering:** The server formats JSON responses which are rendered dynamically in the browser dashboard layout.

5. Advantages of the Proposed Architecture

- **Modular Development:** Code is divided into clear directories (models, routes, controllers) ensuring high maintainability.
- **Secure Operations:** Session limits, password encryption, and API access checks protect records.
- **Scalability:** Stateless Node endpoints can scale horizontally on server instances.

6. Future Enhancements

- Camera scanning integrations for food barcodes.
- Mobile-specific app builds (React Native).
- Direct hardware syncing from wearable health bands.